

Custodian backend — final report

October 2026. Everything in one place: what was built, what was tested, every use case and its state, why some are unfinished, costs, hosting, and what is left. Written in plain words.

1 • What Custodian is

An email assistant. A person connects their mailbox. Custodian reads every email, decides what it is (a person asking for something, a newsletter, a receipt, spam that is really important), labels it, and shows a short list of what needs attention.

It can clean out bulk mail, unsubscribe from senders, set reminders, track what people asked of you and what you promised, draft replies in your own writing style, and answer questions you type.

Two promises run through all of it:

- **Nothing is ever deleted.** "Clear" moves mail to the provider's trash, where it can be recovered.
- **Nothing permanent happens without a yes.** Before clearing, unsubscribing or sending, the person sees the exact action ("trash 360 messages from LinkedIn") and must confirm it. The server refuses otherwise.

The backend is the server side and does all the thinking. The frontend (the screens) is built separately and talks to the backend.

2 • What was built

PART	WHAT IT DOES, SIMPLY
Mail connection (IMAP)	Logs in to the mailbox and reads, labels, archives, moves to trash, sends, undoes
The pipeline	16 steps every email goes through to decide what it is. Mostly fixed rules; AI only where rules can't decide
Database	Stores what Custodian knows about the mail (not the mail itself). Each customer has their own private section
Accounts	Sign up, confirmation email, log in, log out, reset password
API	92 routes: one for everything the screens need
The gate	Every permanent action needs the person's confirmation of the exact wording
Per-user mailboxes	Each user's mailbox password is stored encrypted; each user's requests open only their own mailbox
The worker	Runs all day with nobody watching. Checks every mailbox every 20 seconds, reads and labels new mail. Can never send or delete

PART	WHAT IT DOES, SIMPLY
The chat	The person types a sentence and the app does the right thing. Every feature can be reached by typing
Subscriptions	Free trial, confirm before charging, change plan, pause if a payment fails, cancel
Handover	Handover document, personal data removed from the code, clean copy ready

How the chat works

1. The person types, for example, "has Ali replied to me?"
2. Gemini (Google's AI) reads the sentence with names hidden and says which feature it means: "check replies, person Ali". That is all Gemini does.
3. Our own code looks it up in our database and writes the answer: "No. You wrote to Ali on Tuesday and nothing has come back."

For a question about what an email says ("what did Ali ask me to do?"), the email text goes to Gemini with names hidden, and the answer names the email it came from. If the answer is not in the mail, the chat says so instead of guessing. The chat never clears, unsubscribes or sends by itself: it shows what would happen and waits for the yes, same as a button.

Gemini was chosen for the chat because it costs about \$0.03 per user per month and needs no training. Running our own model would need a few hundred labelled example sentences first.

3 • Mail providers: what is actually tested

PROVIDER	STATE	WHY
Gmail	Working and tested	47 operations proven on a real mailbox: read every folder, label, archive, trash, rescue from spam, mark read, send, draft, forward, undo. Inbox count unchanged after the test
Outlook / Hotmail	Not working yet	Microsoft stopped accepting passwords in 2022. It needs a sign-in token, and for that the app must be registered with Microsoft once. Only a company Microsoft account can register it; personal accounts are refused (tried twice). The code is written and waits for the company's registration
Yahoo	Written, not tested	Uses the same standard as Gmail, so it should work. No real Yahoo account has been connected
Zoho	Written, not tested	Same
iCloud	Written, not tested	Same
Any other provider	Written, not tested	The server is worked out from the email address automatically

4 • All 50 use cases

40 built and tested, 10 partly done, none untouched. "Built" means the backend logic exists and was run on a real Gmail mailbox. The screens are the frontend's work.

Getting an account

#	USE CASE	WHAT IT MEANS	STATE	HOW
1	Register Account	Make an account with email and password	Built	A real confirmation email is sent and its link confirms the account. Tested
2	Sign In	Log in; signing in never gives mailbox access by itself	Partly	Email and password work. "Sign in with Google / Microsoft" not built
3	Start a Subscription	Card first, 7-day trial, cleanup locked until paid	Partly	All the rules work. No real payment company connected
47	Where My Data Is Stored	Show which region; move on request	Partly	Region shown, move request recorded. The move itself is done by hand
48	Change Plan	Upgrade now, downgrade at the next billing date	Partly	Rules work. No real payment company
49	Cancel Subscription	Runs to end of paid period; nothing in mailbox undone	Partly	Rules work. No real payment company

Connecting a mailbox

#	USE CASE	WHAT IT MEANS	STATE	HOW
4	Gmail by App Password	16-character app password, connect over IMAP	Built	The default connection. Tested
5	Outlook by Microsoft Sign-In	Sign in on Microsoft's page, get a token	Partly	Code written. Waiting on Microsoft registration
6	Yahoo by App Password	Same as Gmail, Yahoo's steps	Partly	Written, not tested
7	Zoho by App Password	IMAP switched on in Zoho first	Partly	Written, not tested
8	iCloud by App-Specific Password	abcd-efgh-ijkl-mnop style password	Partly	Written, not tested
9	Any Other IMAP Mailbox	Work out the server from the address	Built	From the address ending, then the domain's mail records
10	Establish Mailbox Connection	Check the password, find folders, note what's allowed	Built	Checks the password for real before storing it, encrypted
11	Gmail by Google Sign-In	Google's own door	Built	Optional second door. Public use needs Google's paid security review

#	USE CASE	WHAT IT MEANS	STATE	HOW
12	Upgrade to Google Sign-In	Switch doors without losing anything	Built	
13	Disconnect a Mailbox	Erase our copy, list what is lost	Built	Mailbox itself untouched
50	Identify the Mail Provider	Which company runs this address	Built	

The cleanup

#	USE CASE	WHAT IT MEANS	STATE	HOW
14	Scan Mailbox History	Read all old mail headers, resumable	Built	Progress shown: "read 1,200 of 7,300"
15	Review and Clear the Pile	Bulk senders grouped with reasons	Built	Trash or archive, after confirmation. Test mailbox: 6,683 messages in 91 groups
16	Clear a Brand's Mail	A company's many addresses as one	Built	Adverts cleared, receipts kept. Test mailbox: 68 companies
17	Unsubscribe	One-click unsubscribe, then watch	Built	Checks for 14 days whether they really stopped
18	Reclaim Storage Space	Largest messages, grouped	Built	Protected senders marked
46	The First Cleanup	Unsubscribe, pile, brands, space in order	Built	One confirmation each

New mail every day

#	USE CASE	WHAT IT MEANS	STATE	HOW
19	Prioritise New Mail	Person or machine, what they want, labelled	Built	Every decision stored with its reason
20	Recover Mail from Spam	Find real mail wrongly in spam	Built	Never moved automatically
21	Inspect a Suspicious Message	Text only, real link destinations	Built	
22	Track Unanswered Outgoing Mail	Your questions nobody answered	Built	
23	Commitments from Structured Mail	Bookings, orders, invoices become dates	Built	
24	Deadline Warnings	Reminder ahead of a deadline	Built	
33	Requests and Their Deadlines	What was asked, of whom, by when	Built	Test mailbox: 12 requests found
36	Resurface Unread Important Mail	One line, once	Built	

Replying

#	USE CASE	WHAT IT MEANS	STATE	HOW
25	Generate a Reply Draft	Draft in your own style	Built	Names hidden from the AI; sent only after yes
26	Send a Quick Answer	Three fixed short answers	Built	No AI used
35	Forward a Batch	Up to 25 messages as drafts	Built	Recipient only from what the user types
42	Tune My Writing Voice	See and change how drafts sound	Built	Learned from sent mail

Seeing your mail

#	USE CASE	WHAT IT MEANS	STATE	HOW
27	Unified Inbox	All mailboxes in one list	Built	
28	Search	Across all mailboxes	Built	Sender, subject, date. Not inside message text
39	Catch Up After Time Away	Needs you / answered / expired / info	Built	
40	Where a Conversation Stands	Who asked what, who waits on whom	Built	
41	Everything About One Person	Their addresses, what is owed both ways	Built	
43	Work Through Today	One ordered list	Built	

Control and trust

#	USE CASE	WHAT IT MEANS	STATE	HOW
29	Correct a Decision	Overrule any decision	Built	
30	Undo an Action	Everything reversible, log in plain words	Built	
31	Weekly Digest	One summary on your schedule	Built	Nothing sent when there's nothing to say
32	Export or Erase My Data	Everything we hold, or nothing	Built	
34	Connect a Calendar	Read today, add dates after "add this?"	Partly	Google Calendar only
37	Follow-Up Reminder	Remind without moving the message	Built	
38	Rule in Plain English	One sentence becomes a rule	Built	Interpretation shown back before saving
44	Mark a Person Important	Never cleared, always on top	Built	

#	USE CASE	WHAT IT MEANS	STATE	HOW
45	Track What You Promised	"I'll send it Friday" from sent mail	Built	

5 • Why 10 use cases are only partly done

None of them is waiting on more coding from our side alone. Each one needs something from outside.

USE CASES	WHAT IS MISSING	WHAT IT NEEDS
3, 48, 49 Subscription, change plan, cancel	Taking real money	A payment company (e.g. Stripe) chosen and an account opened. All the rules are already built and tested; connecting the company is three functions to fill in
5 Outlook	A Microsoft app registration	The company's Microsoft 365 admin, about 15 minutes, free. Only the "client ID" is needed back
6, 7, 8 Yahoo, Zoho, iCloud	A real test	One test account for each. About an afternoon of testing
2 Sign in with Google / Microsoft	The sign-in button for the account itself	A Google and Microsoft login setup for the website. Email and password already work
34 Calendar	Calendars other than Google	Outlook and Apple calendars use a different standard; not started
47 Data region	Moving the data	Done by hand by whoever runs the servers: copy, check, switch, erase old copy

6 • How everything was tested

Every test runs against real systems (the real database, a real Gmail mailbox, real emails) and cleans up after itself. Each one can be re-run.

TEST	WHAT IT PROVES
IMAP audit	47 mailbox operations on a real Gmail, inbox count unchanged
Accounts test	Sign-up → confirmation → login → logout → reset, 14 steps
Email test	A real confirmation email arrives in about 10 seconds and its link works
Mailbox test	Each account opens only its own mailbox; a second account is refused
API test	Every permanent action is refused without the confirmation
Chat test	22 real sentences, every feature, all answered correctly

TEST	WHAT IT PROVES
Subscription test	The whole life of a subscription, 21 checks: trial, lock, confirm, upgrade, downgrade, failed payment, pause, restore, cancel, end
Worker	One full pass reading and labelling mail

7 • Costs

All figures are estimates from real measurements on one laptop and one test mailbox, then scaled up. Treat them as within a factor of two.

The AI models

MODEL	JOB	RUNS ON	SIZE	SPEED PER EMAIL
Classifier	What kind of email, does it need a reply	Our server	550 MB, 1.7 GB memory	66 ms
Name spotter	Finds names to hide before anything goes to Google	Our server	1.1 GB, 1 GB memory	69 ms
Gemini Flash-Lite	Reading requests, drafting replies, rules, chat	Google	none of ours	1–3 seconds

About 70% of email is handled by rules alone, with no AI.

One user, one day (assumed)

ITEM	VALUE
Emails received	120
Handled by rules only	84
Through our own models	36
Sent to Gemini	26 calls
Database growth	120 KB (measured: 7,391 emails = 7.5 MB)

By number of users, per month

USERS	SERVERS	MEMORY	DATABASE	GEMINI COST	TOTAL
10	1	8 GB	0.2 GB	\$0–2	€15–25
100	1	8 GB	2 GB	\$20	€45–60
1,000	3	40 GB	20 GB	\$190	€300–400
10,000	24	380 GB	200 GB	\$1,900	€2,800–3,500

About €0.30 per user per month at scale.

Using an AI service vs running AI ourselves

MODEL	CHEAPER OPTION	WHY
Classifier and name spotter	Run ourselves	They fit on the server we already need, costing nothing extra. Renting them costs about \$48 a month even when idle. The name spotter must stay with us anyway: its job is to hide private details before anything leaves
Gemini (reading, drafting, chat)	Use Google's service	Running our own equivalent needs a graphics server at about \$580 a month, used or not. Google's service is cheaper until roughly 20,000 users

USERS	EVERYTHING AS A SERVICE	EVERYTHING OURSELVES	MIXED (CHOSEN)
10	€60	€560	€15
100	€85	€570	€50
1,000	€400	€850	€350
10,000	€2,900	€4,500	€2,900

Cheapest way to start

ITEM	CHOICE	COST
Server	One 4-core, 8 GB server running everything	about €11 a month
Database	Free tier of a hosted Postgres	€0 to start
Gemini	Free tier	€0 to start
Confirmation emails	Free tier of an email service	€0
Total	about €11–15 a month until roughly 25 users	

8 • Hosting

- **Today:** nothing is deployed. Everything runs on a laptop, with the database on a hosted Postgres service.
- **To go live:** one small server running the API and the worker, with the database in the same region as the server so they are close to each other. Where to host is the company's choice.
- **One thing measured:** with the database far from the server, a single page could wait 1 to 3 seconds on the network. Putting both in the same region removes this.
- **At larger scale:** the worker checks each mailbox every 20 seconds. Push notifications from the providers, where they offer them, would cut the number of servers needed at 1,000+ users.

9 • What is left

1. **Connect a payment company** for subscriptions.
 2. **Deploy** to a server.
 3. **Microsoft registration** for Outlook, then test it.
 4. **Test Yahoo, Zoho, iCloud** with one real account each.
 5. **Sign in with Google / Microsoft** for the account.
 6. **Search inside message text** (today: sender, subject, date).
 7. **Calendars other than Google.**
 8. **Google's security review**, only if the optional Google sign-in door is offered to the public (Google charges for it, several thousand dollars a year). The default IMAP connection does not need it.
-

10 • Safety rules the system keeps

- Nothing is ever deleted from a mailbox. Trash is a move.
 - Nothing permanent happens without the person confirming the exact wording. This is enforced by the server, not just the screen.
 - Names and private details are hidden before any text goes to Google.
 - Bank, doctor, lawyer and HR emails are never opened at all.
 - What the person types is an instruction; what an email says is only data. An email cannot instruct the system.
 - Each customer's data sits in its own private section of the database.
 - Passwords are never stored readable. Mailbox passwords are encrypted with a key that is not in the database.
 - The worker and the chat never do anything permanent on their own.
-

11 • Where everything is

- **Code:** the GitHub repository, plus a clean copy with no history for the company.
- **Handover document:** `HANDOVER.md` — what was personal and must be recreated under company accounts, how to run and check everything in an hour, what is unfinished.
- **Route contract for the frontend:** `docs/API.md` .
- **Design document:** `docs/Custodian-Backend-Design.pdf` .